Linear Equations & Systems

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the botanical conservatory, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?

Answer: 1. From the graph in the xy-plane, line ℓ_1 passes through $(0, 1)$ and $(1, 2)$, and line ℓ_2 passes through $(0, 4)$ and $(1, 2)$. The two lines intersect at the point $(1, 2)$. Therefore, the required value is 1.

2 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the regional bakery, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?

Answer: 4. From the graph in the xy-plane, line ℓ_1 passes through $(0, -5)$ and $(4, 3)$, and line ℓ_2 passes through $(0, 7)$ and $(4, 3)$. The two lines intersect at the point $(4, 3)$. Therefore, the required value is 4.

3 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the science museum, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?

Answer: 2. From the graph in the xy-plane, line ℓ_1 passes through $(0, 2)$ and $(2, 4)$, and line ℓ_2 passes through $(0, 8)$ and $(2, 4)$. The two lines intersect at the point $(2, 4)$. Therefore, the required value is 2.

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4 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the ceramic studio, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x ?

Answer: 5. From the graph in the xy-plane, line ℓ_1 passes through $(0, -5)$ and $(5, 5)$, and line ℓ_2 passes through $(0, 10)$ and $(5, 5)$. The two lines intersect at the point $(5, 5)$. Therefore, the required value is 5.

5 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the public transit depot, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?

Answer: 3. From the graph in the xy-plane, line ℓ_1 passes through $(0, 3)$ and $(3, 6)$, and line ℓ_2 passes through $(0, 12)$ and $(3, 6)$. The two lines intersect at the point $(3, 6)$. Therefore, the required value is 3.

6 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the art gallery, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?

Answer: 1. From the graph in the xy-plane, line ℓ_1 passes through $(0, 0)$ and $(1, 2)$, and line ℓ_2 passes through $(0, 3)$ and $(1, 2)$. The two lines intersect at the point $(1, 2)$. Therefore, the required value is 1.

7 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the university lecture hall, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?

Answer: 4. From the graph in the xy-plane, line ℓ_1 passes through $(0, -1)$ and $(4, 3)$, and line ℓ_2 passes through $(0, 11)$ and $(4, 3)$. The two lines intersect at the point $(4, 3)$. Therefore, the required value is 4.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

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8 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the orchestra hall, $\{6x + y = -3;\ 18x + ky = 7$ For what value of k does this system of equations have no solution?

Answer: 3. The lines are parallel (no solution) when the coefficients are proportional: $\frac{18}{6} = \frac{k}{1} \Rightarrow k = 3$ (and the constants are not in that same ratio).

9 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the honeybee apiary, a theater sold 29 tickets for a total of USD 310. Adult tickets cost USD 16 and child tickets cost USD 9. How many adult tickets were sold?

Answer: 7. Let a = adult tickets. $16a + 9(29 - a) = 310 \Rightarrow 7a = 49 \Rightarrow a = 7$.

10 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the local farmers market, $\{6x - 3y = 0;\ 5x - 6y = 28$ What is the value of x?

Answer: -4. Solving the system by elimination gives $x = -4$ and $y = -8$.

11 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the performing arts academy, the real number t satisfies $\frac{23t + 20}{2} = 391 + \frac{20}{2}$. What is the linear expression $p = 23t - 20$?

Answer: 762. Multiplying by 2 gives $23t = 782$, so $23t - 20 = 782 - 20 = 762$.

12 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the coastal aquarium, the linear system $17t + 6y = 284$ and $34t + (p + 6)y = 578$ has no solution. What is the constant parameter p?

Answer: 6. A linear system has no solution when the lines are parallel and distinct: $\frac{34}{17} = \frac{p + 6}{6} \Rightarrow p = 6$.

13 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the public transit depot, $\{3x + y = 2;\ 6x + ky = 14$ For what value of k does this system of equations have no solution?

Answer: 2. The lines are parallel (no solution) when the coefficients are proportional: $\frac{6}{3} = \frac{k}{1} \Rightarrow k = 2$ (and the constants are not in that same ratio).

14 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the wildlife sanctuary, the ordered pair (t, y) satisfies $t + y = 26$ and $17t + 3y = 302$. What is the coordinate t in $p(t, y)$?

Answer: 16. Substituting $y = 26 - t$ into $17t + 3y = 302$ yields $t = 16$.

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Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the historic lighthouse, if the linear equation $6(x+4)=6x+c$ has infinitely many solutions for x, what is the value of the constant c?

Answer: 24. Expanding the left side gives $6x+24=6x+c$. For the equation to hold for all real x, the constant terms must be equal, so $c=24$.

16 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the botanical conservatory, adult tickets sell for USD 5 each and student tickets sell for USD 8 each. On Saturday, a total of 18 of these items were sold for a combined revenue of USD 114. How many adult tickets were sold on Saturday?

Answer: 10. Let a be the number of adult tickets and b be the number of student tickets. Then $a+b=18$ and $5a+8b=114$. Substituting $b=18-a$ gives $5a+8(18-a)=114 \Rightarrow -3a=-30 \Rightarrow a=10$.

17 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In the system of linear equations $3x+4y=11$ and $9x+ky=26$ modeling two supply lines at the municipal library, k is a constant. If the system has no solution, what is the value of k?

Answer: 12. A system of two linear equations has no solution when the lines are parallel and have different y-intercepts. Multiplying the first equation's x-coefficient by 3 gives 9, so $k=4\times 3=12$.

18 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

For the community theater inventory model, the equation $\frac{1}{2}x+4=\frac{42}{6}$ relates the reserve count x to the target quota. What is the value of x?

Answer: 6. Multiply both sides of the equation by 6: $3x+24=42 \Rightarrow 3x=18 \Rightarrow x=6$.

19 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

For a calibration balance at the regional bakery, if the linear equation $6(x+7)=6x+c$ has infinitely many solutions for x, what is the value of the constant c?

Answer: 42. Expanding the left side gives $6x+42=6x+c$. For the equation to hold for all real x, the constant terms must be equal, so $c=42$.

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20 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the coastal aquarium, oak saplings sell for USD 9 each and maple saplings sell for USD 12 each. On Saturday, a total of 24 of these items were sold for a combined revenue of USD 246. How many oak saplings were sold on Saturday?

Answer: 14. Let a be the number of oak saplings and b be the number of maple saplings. Then $a + b = 24$ and $9a + 12b = 246$. Substituting $b = 24 - a$ gives $9a + 12(24 - a) = 246 \Rightarrow -3a = -42 \Rightarrow a = 14$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the municipal library, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of y ?

a. 6

b. 3

c. 9

d. 4 ✓

From the graph in the xy-plane, line ℓ_1 passes through $(0, 0)$ and $(2, 4)$, and line ℓ_2 passes through $(0, 6)$ and $(2, 4)$. The two lines intersect at the point $(2, 4)$. Therefore, the required value is 4.

22 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the community theater, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of $x + y$?

a. 8

b. 19

c. 9 ✓

d. 11

From the graph in the xy-plane, line ℓ_1 passes through $(0, 3)$ and $(3, 6)$, and line ℓ_2 passes through $(0, 12)$ and $(3, 6)$. The two lines intersect at the point $(3, 6)$. Therefore, the required value is 9.

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23 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the coastal aquarium, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of y?

a. 5 ✓

b. 7

c. 4

d. 26

From the graph in the xy-plane, line ℓ_1 passes through $(0, 0)$ and $(5, 5)$, and line ℓ_2 passes through $(0, 15)$ and $(5, 5)$. The two lines intersect at the point $(5, 5)$. Therefore, the required value is 5.

24 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the mountain observatory, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of $x + y$?

a. 5

b. 2

c. 6

d. 3 ✓

From the graph in the xy-plane, line ℓ_1 passes through $(0, 0)$ and $(1, 2)$, and line ℓ_2 passes through $(0, 3)$ and $(1, 2)$. The two lines intersect at the point $(1, 2)$. Therefore, the required value is 3.

25 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the urban bike-share station, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of y?

a. 19

b. 6 ✓

c. 8

d. 5

From the graph in the xy-plane, line ℓ_1 passes through $(0, 0)$ and $(3, 6)$, and line ℓ_2 passes through $(0, 9)$ and $(3, 6)$. The two lines intersect at the point $(3, 6)$. Therefore, the required value is 6.

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26 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the solar microgrid facility, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of $x + y$?

a. 7 ✓

b. 9

c. 6

d. 13

From the graph in the xy-plane, line ℓ_1 passes through $(0, -1)$ and $(4, 3)$, and line ℓ_2 passes through $(0, 11)$ and $(4, 3)$. The two lines intersect at the point $(4, 3)$. Therefore, the required value is 7.

27 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the agricultural cooperative, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of y?

a. 1

b. 3

c. 2 ✓

d. 4

From the graph in the xy-plane, line ℓ_1 passes through $(0, 1)$ and $(1, 2)$, and line ℓ_2 passes through $(0, 4)$ and $(1, 2)$. The two lines intersect at the point $(1, 2)$. Therefore, the required value is 2.

28 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the publishing house, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of $x + y$?

a. 9

b. 6 ✓

c. 8

d. 5

From the graph in the xy-plane, line ℓ_1 passes through $(0, 0)$ and $(2, 4)$, and line ℓ_2 passes through $(0, 6)$ and $(2, 4)$. The two lines intersect at the point $(2, 4)$. Therefore, the required value is 6.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

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29 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the wildlife sanctuary, a store sells notebooks for USD 3 each and pens for USD 1.50 each. Maria spent USD 18 on a total of 8 items. How many notebooks did she buy?

a. 2

b. 3

c. 4 ✓

d. 5

Let n = notebooks: $3n + 1.5(8 - n) = 18 \Rightarrow 1.5n = 6 \Rightarrow n = 4$.

30 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the charter ferry service, in the plane, the line $14x + 15y = 210$ intersects the axes at points A and B. What is the sum of intercepts f of $14x + 15y = 210$?

a. 29 ✓

b. 43

c. 31

d. 14

The intercepts are 15 and 14, whose sum is $15 + 14 = 29$.

31 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the textile workshop, in a calculation for the community orchard (case 44), what is the sum of all solutions to the equation $|2x - 10| = 4$?

a. 17

b. 10 ✓

c. 7

d. 16

$2x - 10 = 4$ gives $x = 7$; $2x - 10 = -4$ gives $x = 3$. Sum $= 10$.

32 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the geology lab, the ordered pair (x, y) satisfies $x + y = 16$ and $20x + 4y = 240$. What is the coordinate x in $f(x, y)$?

a. 7

b. 13

c. 31

d. 11 ✓

Substituting $y = 16 - x$ into $20x + 4y = 240$ yields $x = 11$.

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33 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the carpentry workshop, in the coordinate plane, consider the lines $f(x) = 6x - 14$ and $g(x) = 76$. What is the intersection coordinate x where $f(x) = g(x)$?

a. 21

b. 17

c. 15 ✓

d. 1

Setting $6x - 14 = 76$ yields $6x = 90 \Rightarrow x = 15$.

34 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the historic lighthouse, $\{6x + 6y = -8; 12x + ky = -7$ For what value of k does this system of equations have no solution?

a. 12 ✓

b. 14

c. 11

d. 21

The lines are parallel (no solution) when the coefficients are proportional: $\frac{12}{6} = \frac{k}{6} \Rightarrow k = 12$ (and the constants are not in that same ratio).

35 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the community orchard, the point $A(19, 14)$ lies on the line $fx - 4y = 134$. What is the coefficient f?

a. 12

b. 10 ✓

c. 20

d. 6

Substituting $(19, 14)$ gives $19f = 190 \Rightarrow f = 10$.

36 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the environmental club, a linear function $p(t)$ satisfies $p(0) = 9$ and $p(7) = 58$. What is the input t for which $p(t) = 100$?

a. 20

b. 15

c. 4

d. 13 ✓

The slope is $m = 7$, so $p(t) = 7t + 9$. Setting $7t + 9 = 100$ gives $t = 13$.

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37 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the neighborhood hardware store, $\{6x - 6y = 54; 2x + 3y = -2$ What is the value of $x - y$?

a. 0

b. 9 ✓

c. 16

d. 11

Solving the system by elimination gives $x = 5$ and $y = -4$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the honeybee apiary, in the system of linear equations $2x + 7y = 35$ and $6x + ky = 50$, k is a constant. If the system has no solution, what is the value of k?

a. 10

b. 6

c. 23

d. 21 ✓

A system of two linear equations has no solution when the lines are parallel and have different y-intercepts. Multiplying the first equation's x-coefficient by 3 gives 6, so $k = 7 \times 3 = 21$.

39 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the carpentry workshop, for the textile workshop inventory model, the equation $\frac{1}{2}x + 7 = \frac{87}{6}$ relates the reserve count x to the target quota. What is the value of x?

a. 13

b. 22

c. 15 ✓

d. 17

Multiply both sides of the equation by 6: $3x + 42 = 87 \Rightarrow 3x = 45 \Rightarrow x = 15$.

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40 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the community orchard, at the geology lab, soccer balls sell for USD 9 each and training cones sell for USD 12 each. On Saturday, a total of 26 of these items were sold for a combined revenue of USD 270. How many soccer balls were sold on Saturday?

a. 14 ✓

b. 12

c. 16

d. 24

Let a be the number of soccer balls and b be the number of training cones. Then $a + b = 26$ and $9a + 12b = 270$. Substituting $b = 26 - a$ gives $9a + 12(26 - a) = 270 \Rightarrow -3a = -42 \Rightarrow a = 14$.

41 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the local farmers market, in the system of linear equations $6x + 4y = 39$ and $18x + ky = 54$, k is a constant. If the system has no solution, what is the value of k?

a. 7

b. 18

c. 14

d. 12 ✓

A system of two linear equations has no solution when the lines are parallel and have different y-intercepts. Multiplying the first equation's x-coefficient by 3 gives 18, so $k = 4 \times 3 = 12$.

42 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the neighborhood bookstore, wooden chairs sell for USD 9 each and folding tables sell for USD 12 each. On Saturday, a total of 30 of these items were sold for a combined revenue of USD 318. How many wooden chairs were sold on Saturday?

a. 14 ✓

b. 16

c. 28

d. 17

Let a be the number of wooden chairs and b be the number of folding tables. Then $a + b = 30$ and $9a + 12b = 318$. Substituting $b = 30 - a$ gives $9a + 12(30 - a) = 318 \Rightarrow -3a = -42 \Rightarrow a = 14$.

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43 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In the system of linear equations $4x + 6y = 27$ and $12x + ky = 42$ modeling two supply lines at the robotics club, k is a constant. If the system has no solution, what is the value of k?

a. 9

b. 12

c. 20

d. 18 ✓

A system of two linear equations has no solution when the lines are parallel and have different y-intercepts. Multiplying the first equation's x-coefficient by 3 gives 12, so $k = 6 \times 3 = 18$.

44 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

For the university lecture hall inventory model, the equation $\frac{1}{2}x + 6 = \frac{57}{6}$ relates the reserve count x to the target quota. What is the value of x?

a. 5

b. 13

c. 7 ✓

d. 9

Multiply both sides of the equation by 6: $3x + 36 = 57 \Rightarrow 3x = 21 \Rightarrow x = 7$.

45 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

For a calibration balance at the culinary institute, if the linear equation $4(x + 5) = 4x + c$ has infinitely many solutions for x, what is the value of the constant c?

a. 24

b. 20 ✓

c. 9

d. 5

Expanding the left side gives $4x + 20 = 4x + c$. For the equation to hold for all real x, the constant terms must be equal, so $c = 20$.

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